

Regional Development of Transport Infrastructure and Communications – Course Assignments Summary

 Deadline for all assignments: January 11, 2025 (23:59, Moodle submission)

Each student must complete and submit three assignments that progressively build skills in data use, GIS mapping, spatial analysis, and critical thinking. Assignments are ordered below from the easiest (introductory) to the most advanced (research project).

Assignment 1 – Airport Flow Map (Introductory – Practical GIS Task)

Title: Global Connectivity of an International Airport

Objective

The objective of this task is to create a **flow map** illustrating the spatial connectivity of one selected **international airport** with its regular flight destinations. Through this task, students will apply **geographical visualization** and **network analysis** concepts to represent global air mobility patterns and spatial interaction intensity.

Rationale

Airports are among the most significant **nodes in global transport networks**, linking regions, economies, and people. Flow maps help visualize **the magnitude and direction of connections**, providing insight into the **hierarchy, reach, and spatial organization** of air transport systems. This assignment combines **GIS mapping, data sourcing, and cartographic design** to analyze these interactions.

Instructions

Step 1. Select an Airport

Choose **one international airport** of your choice. Make sure it has sufficient and recent data on **regular flight connections and destinations**.

Examples:

- Heathrow (LHR), Frankfurt (FRA), Istanbul (IST), Sofia (SOF), Dubai (DXB), Doha (DOH), or Singapore (SIN).
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Step 2. Data Sources

Use **open and freely available datasets**:

- [Global Airports – Humanitarian Data Exchange \(HDX\)](#)
- [World Airports – ArcGIS Hub](#)

Complement this with:

- The **official website** of your chosen airport (for flight destinations).
 - Optionally, consult **OpenFlights.org** or **FlightRadar24 (free mode)** for route verification.
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Step 3. Create the Flow Map

- Watch the tutorial video: [Flow Map Tutorial – Made4Geek](#)
- Use **ArcGIS Pro**, **QGIS**, or another GIS platform capable of flow mapping.
- Display lines connecting your chosen airport to its regular international destinations.
- You may vary **line thickness** to represent:
 - Flight frequency,
 - Passenger volume, or
 - Distance (if data available).

Tip: Use the airport's IATA or ICAO codes to link the attribute tables between origin and destination points.

Step 4. Output Format

- Create the final map as a **JPEG file** (high-resolution, A3 format preferred). Include:
 - A clear title,
 - Legend and scale bar,
 - Projection information,
 - Short caption or note on data sources.

 **Deadline:** Upload your JPEG file to Moodle **by January 11, 2025**.

Assessment Criteria (100 points)

Criterion	Description	Points
Data accuracy	Correct and complete set of destinations, up-to-date data	25
Cartographic quality	Clarity, layout, labeling, color harmony, and overall readability	25
Spatial insight	Ability to interpret and represent spatial connectivity	20
Use of GIS tools	Appropriate techniques for data handling, symbology, and projection	20
Deadline compliance	Timely submission and correct format	10

Additional Suggestions

- Include a **small inset map** showing the airport's location within its country or region.
- Optionally, classify destinations by **continent or region** using color coding.
- For advanced students: calculate **mean flight distance** or **directional distribution (standard deviational ellipse)** to show network dispersion.

Expected Learning Outcomes

By completing this task, you will:

1. Apply **network and spatial interaction concepts** to real-world air transport systems.
2. Gain hands-on experience with **GIS visualization and data integration**.
3. Understand how **flow maps reveal functional connectivity** and global linkages in transport geography.

Final Submission Checklist

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|--|---|
| ☒ High-resolution JPEG map | ☒ Data sources cited |
| ☒ Proper labeling and legend | ☒ Uploaded to Moodle before 11.01.25 |

Assignment 2 – Logistics and Network Poster (Intermediate – Applied & Analytical Task)

Title: Mapping the Efficiency and Vulnerability of Urban Supply Networks: A GIS-Based Poster

 Aim To apply theoretical concepts from transport geography and logistics (network topology, derived demand, and cross-docking) to a real-world urban context, using GIS tools to analyze, visualize, and interpret spatial patterns of connectivity, accessibility, and supply chain vulnerability.

Learning Objectives

By completing this assignment, you will be able to:

1. Apply **geographical thinking** to understand spatial organization and interdependencies in transport systems.
2. Use **GIS-based spatial analysis** to model connectivity, accessibility, and flow between key nodes (e.g., warehouses, retail stores, ports).
3. Identify and interpret **bottlenecks, redundancies, and spatial inequalities** in supply networks.
4. Synthesize results in a **scientific poster** that communicates data, analysis, and implications for sustainable urban logistics.

Task Description

Students will:

1. **Select one major city** in Europe (e.g., Sofia, Prague, Warsaw, Milan, Budapest etc.).
2. **Map the main logistics network** of the city, including:
 - Major **transport nodes** (ports, airports, warehouses, intermodal terminals).
 - **Transport links** (roads, railways, main corridors).
 - **Retail or consumption nodes** (supermarkets, distribution centers).
3. Using **GIS tools (ArcGIS Pro, QGIS, or Google Earth Engine)**:
 - Conduct a **network analysis** (e.g., shortest path or service area) to identify efficient vs. congested routes.
 - Perform a **centrality analysis** (degree, closeness, betweenness) on transport nodes using open data (e.g., OpenStreetMap or Geofabrik shapefiles).

- Optionally overlay **population density or land use data** to link logistics accessibility with urban structure.
 - 4. Reflect on **what would happen if one key node fails** (e.g., port closure, bridge damage, or labor shortage).
 - Simulate the impact on network connectivity.
 - Suggest alternative routing or cross-docking strategies.
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Recommended Free Data Sources

- **OpenStreetMap (OSM)** – base map & transport networks (<https://download.geofabrik.de/>)
 - **Eurostat GISCO** – administrative boundaries, population data
 - **Copernicus Open Access Hub** – satellite imagery (optional for land-use analysis)
 - **Google Earth Engine or QGIS plugins** for routing or network modeling
 - **National open data portals** (e.g., data.gov.bg, data.gov.uk)
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Expected Outputs

1. A **GIS-based analysis** (maps and layers) showing:
 - Transport network topology
 - Key nodes and flows
 - Accessibility or centrality visualization
 - Potential vulnerabilities
2. A **poster (A1 format)** summarizing:
 - Introduction (concept and study area)
 - Methods (data & GIS tools used)
 - Results (maps, graphs, findings)
 - Discussion: spatial implications and sustainability insights
 - References

Deadline

Submit all materials (poster PDF + project folder with GIS files) in Moodle **by January 11, 2025**.

Evaluation Criteria (100 points)

Criterion	Description	Points
Conceptual Understanding	Clear application of transport geography concepts (network, derived demand, cross-docking, etc.)	20
GIS Methodology	Correct and innovative use of GIS tools and open data	25
Analytical Depth	Interpretation of spatial patterns and network logic	20
Visual Communication	Clarity, readability, and aesthetics of the poster	20
Reflection	Insight on sustainability, efficiency, and resilience	15

Bonus Option (up to +5 points)

Include a **short 1-minute audio or video explanation** embedded in your poster QR code, summarizing your findings and visualizing network efficiency or failure scenarios.

Assignment 3 – Course Project (Advanced – Analytical & Research Task)

Title: Development and Current State of the Transport Infrastructure of a Selected Agglomeration

The aim of this course project is to analyze the **historical development, current structure, and future perspectives** of the transport infrastructure of a selected **urban agglomeration**. You will apply **geographical thinking** and **systemic analysis** to understand how transport networks evolve, interact with urban growth, and contribute to regional and global connectivity.

Task Description

Each student will:

1. **Select an agglomeration** (preferably a European or global metropolitan area with diverse transport systems — e.g. London, Paris, Warsaw, Prague, Milan, Sofia, Istanbul, Bucharest, etc.).
2. **Research and analyze:**
 - The historical evolution of transport infrastructure
 - The spatial organization of modern transport systems (road, rail, air, public, and sustainable modes)
 - The relationship between transport, urbanization, and sustainable development
3. **Produce a written report (course project)** with maps, figures, tables, and references.
 - Length: **2500–3000 words** (approx. 8–10 pages, excluding bibliography).
 - Format: Word or PDF, academic writing style.
 - Include **at least one original map or spatial visualization** (created in GIS, Google Earth, or other mapping tools).

Suggested Structure (Example with London)

Students may adapt this outline to their chosen city.

1. Introduction

- Relevance of the topic and purpose of the study.
- Importance and role of transport infrastructure in urban development.

- The city's function as a **global, regional, or national hub**.
- Research methods, approaches, and data sources (e.g. official transport plans, statistical databases, GIS data, academic studies).
- Short literature review (2–3 key references).

2. Historical Development of the Transport Infrastructure

- Early transport systems (roads, canals, first railways).
- Industrial expansion and modernization (rail transport, underground networks).
- Post-war developments (motorways, suburban systems).

3. Modern Transport Infrastructure

- **3.1. Road Transport:** main motorways, congestion policies, low-emission zones.
- **3.2. Public Transport:** metro, bus, tram, and suburban rail systems.
- **3.3. Air Transport:** airports, connectivity, and passenger flows.
- **3.4. Non-Motorized and Sustainable Transport:** cycling, walking networks, shared mobility.
- **3.5. Intelligent and Integrated Systems:** digital technologies, smart cards, apps, and real-time data systems.

4. Problems and Challenges

- Congestion and overcrowding.
- Aging infrastructure and maintenance.
- Environmental impacts and emissions.
- Urban growth, population pressure, and planning issues.

5. Future Development and Prospects

- Ongoing and planned major projects (e.g. new metro lines, high-speed rail, or airport expansion).
- Sustainable mobility initiatives (EV infrastructure, green corridors).
- Urban planning integration and policy vision.

6. Recommendations

- Proposals for improving efficiency, sustainability, and inclusiveness.

- Balancing transport expansion with environmental and social goals.

7. Conclusion

- Concise summary of key findings and reflections on the future role of transport in the selected agglomeration.

8. Bibliography

- Use consistent academic referencing (APA, Harvard, or Chicago).
 - Include all sources: academic literature, statistical data, GIS portals, and official policy documents.
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Expected Learning Outcomes

After completing this coursework, you will be able to:

- Demonstrate **understanding of urban transport systems** as complex spatial and functional networks.
 - Apply **geographical analysis** to interpret infrastructure patterns and evolution.
 - Use **cartographic and spatial visualization tools** (GIS, map design).
 - Evaluate **policy and planning dimensions** of transport development in large cities.
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Recommended Data and Tools

- **OpenStreetMap & Geofabrik** – road and rail networks
- **EUROSTAT & Urban Audit** – demographic and transport statistics
- **Copernicus or Sentinel Hub** – land-use and spatial imagery
- **Google Earth / QGIS / ArcGIS Online** – mapping and visualization
- **Official local data portals** (e.g. TfL, INSEE, NSTAT, NSI Bulgaria)

Assessment Criteria (100 points)

Criterion	Description	Points
Relevance & Structure	Logical organization and clarity of sections	15
Depth of Analysis	Quality of discussion on transport systems and evolution	20
Spatial Perspective	Use of maps, GIS, and geographical reasoning	20
Use of Sources	Reliability and diversity of data and references	15
Critical Insight	Reflection on challenges, sustainability, and planning	20
Presentation	Academic language, formatting, citation style	10

Submission Deadline

 Upload the final report (PDF or DOCX) to **Moodle**, including maps and references **by January 11, 2025**.

Overall Learning Progression

-  Beginner – Assignment 1: Flow Map – Basic GIS skills.
-  Intermediate – Assignment 2: Poster – Applied network analysis.
-  Advanced – Assignment 3: Course Project – Full spatial and analytical research.

Submission Guidelines

- All submissions due on Moodle by January 11, 2025 (23:59).
- Each file must include student name, ID, course name, and assignment number.
- File name format: Lastname_Firstname_AssignmentX_January2025.

Grading Overview (100 points per assignment)

- Conceptual understanding and application of theory
- Data accuracy and source reliability
- Quality of maps, visuals, and analysis
- Clarity of writing and layout
- Timely submission